Research Interests. My research interests lie at the intersection of databases, systems, and programming languages. My current focus is on exploring the efficiency and reliability aspects of data systems. More concretely:

1. **State efficient Incremental View Maintenance**: Incremental compute engines present the next frontier of optimization in data systems by trading higher storage for lower latency. However, under complex workloads, this trade-off breaks down as the state grows super-linearly. I plan on exploring partial computation techniques to allow for state-efficient IVM, making it practical for resource-constrained environments.
2. **Query optimization in traditional databases using Incremental View Maintenance**: Recent progress in IVM, including our work on DBSP, has made it possible to support low-latency incremental maintenance for an unrestricted set of queries. However, these advances have not yet been adapted in traditional database systems. Existing efforts, such as `pg_ivm`, remain limited[1]. In contrast, our work at Feldera demonstrates that unrestricted IVM is practical, effective, and general enough to support user-defined functions. I am interested in exploring how these ideas can be integrated into traditional databases, such as PostgreSQL or DuckDB, to simplify complex queries, reduce unnecessary full table scans, and improve the performance of workloads that currently rely on expensive batch re-evaluation.

These topics reflect my current focus, though I remain open to related problems across database systems.

Motivation. My fascination with databases began during my final-year project, when out of curiosity and as a personal challenge, I decided to implement one from scratch. Additionally, in order to avoid the client having to poll the database repeatedly, I implemented a WebSocket server within the database that would send the latest version of the view to the client application directly. This hands-on effort exposed me to the inner workings of data systems, from query optimization to storage engines, and sparked a deep interest that has stayed with me since.

Additionally, working at an open-source data systems startup for the past two years has deepened my interest in databases and given me firsthand exposure to the research and engineering challenges involved in building a novel data system. Together, these experiences have formed the foundation of my long-term goal of becoming a database researcher.

Research Experience. My research focuses on a simple observation: most applications are, at their core, views over databases. Keeping this central principle, my work at Feldera focuses on **automatic incremental view maintenance** (IVM) to make data pipelines more efficient and keep the insights fresh. This work builds on the foundations of differential dataflow to propose DBSP, a simple but expressive language for describing computations over data streams. We then demonstrate how to build upon DBSP to support rich query languages like SQL. This is where my contributions primarily lie, in allowing users to write unrestricted SQL that ultimately compiles down to efficient DBSP circuits. Additionally, to make the system more well-rounded, I contributed to developing connectors that allow seamless integration with popular external sources like Kafka and Postgres. This work led to a special issue publication in the VLDB journal.

When building the SQL interface atop DBSP, we found that relying on Apache Calcite for query optimizations introduced numerous bugs related to the type system and inconsistencies between optimized and unoptimized query results. To address this correctness issue systematically using differential testing, I independently extended SQLancer[2] for Feldera. SQLancer would build pipelines with different tables and define two different versions of a view, based on equivalent queries, where one version

[1] `pg_ivm` is comparatively inefficient and only supports a limited subset of queries; for instance, it lacks support for recursive queries.

[2] Manuel Rigger and Zhendong Su: Detecting optimization bugs in database engines via non-optimizing reference engine construction. In Proceedings of the 28th ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE 2020). Association for Computing Machinery, New York, NY, USA, 1140–1152 (2020). `https://doi.org/10.1145/3368089.3409710`

is easier for the compiler to optimize and another is much harder. In a couple of months, this effort significantly improved the reliability of the compiler by finding over 30 critical internal compiler bugs.

Why PhD? Being part of a founding team composed mostly of PhDs for the last two years has shaped my thinking and taught me to approach problems from their first principles, and critically design solutions before implementing them. I am deeply grateful for this environment, which provided me with the rare opportunity to engage in PhD-level research within an industry setting. I now feel uniquely positioned to pursue a PhD, bringing research experience and technical maturity that will allow me to contribute meaningfully from the outset.

Now, I am seeking to deepen my focus in research, beyond the inevitable context-switching required in industry. Pursuing a PhD is a deliberate choice to prioritize long-term, foundational research and to build toward my goal of becoming an independent database researcher.

University of Texas at Austin. UT Austin is an exceptional fit for my research interests in databases and data systems in general. I am especially excited to work with **Professor Dixin Tang** to explore techniques to reduce the state required for incremental computation. To reduce state usage, ideally, we want to discard all inputs and intermediate values that can no longer affect the query output; however, it is difficult to determine this for data that is not append-only. One promising direction involves improving the state efficiency in incremental computation systems by using leveraging **Professor Tang's work on incrementability**[3] to determine the optimal balance between pre-computation and on-demand computation dynamically.

While this direction reflect where I can contribute immediately, I am equally open to pursuing other problems across databases and data systems.

Conclusion. Having spent the past two years working at the intersection of research and engineering, I have developed the discipline, technical maturity, and research perspective necessary for doctoral work. I am comfortable navigating ambiguity, collaborating, and exploring open-ended problems. What I am now looking for is the depth and continuity that only an academic environment can provide. UT Austin offers exactly this environment and aligns perfectly with my interests in building efficient and reliable data systems.

[3] Tang, D., Shang, Z., Elmore, A.J., Krishnan, S., Franklin, M.J.: Thrifty Query Execution via Incrementability. In: Proceedings of the 2020 ACM SIGMOD International Conference on Management of Data. pp. 1241–1256. ACM, Portland OR USA (2020)